

Dhruv Shah

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EDUCATION

Carnegie Mellon University

B.S. in Electrical and Computer Engineering, Minor in AI

GPA: 3.73, August 2025 – May 2029

Pittsburgh, PA

Relevant Coursework: Digital Systems, Data Structures & Algorithms, Signals and Systems, Discrete Mathematics, Probability

TECHNICAL EXPERIENCE

AI/ML Intern | Netramind Innovations

May 2026 – August 2026

- Developed a deep learning pipeline for ultra-widefield fluorescein angiography segmentation of retinal total area and non-perfusion area, benchmarking UNet, UNet++, MANet, MAE-pretrained, dual-decoder, and CNN-RNN variants; best model achieved 0.963 TA Dice, 0.494 NPA Dice, and 0.879 NPA AUC on a patient-level held-out test set.
- Built end-to-end evaluation and clinical measurement tooling for NPA analysis, including Dice, precision, recall, AUC, pixel-to-mm² area estimation, NPA/TA burden ratios, model comparison tables, prediction overlays, and inference scripts for deployment.
- Designed temporal FA sequence experiments using 624 multiframe images across 70 cases, implementing QC-filtered frame selection, probability-based mask collation, and CNN-RNN architectures with LSTM/ConvGRU fusion to study whether angiography time-series context improves NPA segmentation.

Software Engineering Intern | Lockheed Martin

May 2025 – July 2025

- Developed a Python wrapper for the AI model Stable Diffusion XL (SDXL), integrating REST API endpoints and JSON-based inference requests to expose configurable generation parameters including resolution, guidance scale, seed, and batch size.
- Integrated SDXL into internal generative-AI workflows for automated image and logo generation, while prototyping LangChain-based LLM orchestration and agentic workflows for multimodal AI applications.

Computational Research Assistant | University of Florida

May 2024 – July 2024

- Adapted a FORTRAN-based compressible-flow CFD solver (HyBurn) to model conventional, supercritical, and diamond airfoils using immersed-boundary .STL geometries across Mach 0.5, 0.8, 1.2, and 1.5.
- Analyzed lift/drag coefficients, pressure distributions, and shock-wave behavior across transonic and supersonic regimes; authored a 4,000-word research paper and presented findings through a technical poster and formal presentation.

PROJECTS

Handwritten Digit Recognition | Python, TensorFlow/Keras, OpenCV

- Designed and trained a neural network in TensorFlow/Keras on 60,000+ MNIST images for handwritten digit classification, implementing end-to-end training, validation, and inference pipelines.
- Engineered an OpenCV computer-vision pipeline for real-world handwritten inputs using grayscale conversion, thresholding, contour-based digit localization, resizing, and normalization to transform user-generated images into model-ready tensors.

iOS Application | Swift, iOS

- Designed and developed a production iOS application in Swift, implementing application logic, interactive user interfaces, and end-to-end functionality for deployment through the Apple App Store.
- Managed the full application lifecycle from development and testing through App Store deployment, scaling the application to 200+ downloads across 4 continents, with a 5-star rating.

LEADERSHIP

Orientation Counselor | Carnegie Mellon University

August 2026

- Selected to mentor incoming students during CMU Orientation, leading group of 30 students through programming and coordinating with university staff to support first-year transition.

AWARDS

Dean's List | SAT: 1550/1600 | ACT: 35/36 | 7th Nationally, BPA Economic Research | 13th Nationally, Science Olympiad Elastic-Launched Glider | 11th Statewide, Mu Alpha Theta Calculus Division | Presidential Award for Academic Excellence | National Merit Finalist

SKILLS

Languages: Python, C/C++, Java, JavaScript, Swift, FORTRAN, SystemVerilog

AI/ML: PyTorch, TensorFlow/Keras, OpenCV, scikit-learn, NumPy, Pandas, CNNs, Computer Vision, Semantic Segmentation, Self-Supervised Learning, Diffusion Models, LLMs

Hardware & Embedded: FPGA, RTL Design, Digital Logic, Logic Synthesis, Testbenches, Boolean Board, Raspberry Pi/Pico, Microcontrollers, I²C, GPIO, Sensors & Peripheral Interfacing

Frameworks & Tools: LangChain, LangGraph, REST APIs, Git/GitHub, Linux/Unix, Jupyter, Pillow, Agile/JIRA, Vivado

Languages: English (fluent/native), Spanish (fluent)